

Chanukya Gattu

Senior Software Engineer | Distributed Systems | Backend & Real-Time Streaming Platforms

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SUMMARY

Senior Software Engineer with **10+ years** building distributed backend and real-time data platforms at JPMorgan Chase, Capital One, AT&T, and Verizon, specializing in Java, Spring Boot, Kafka, Flink, and AWS. At JPMorgan, own the API platform for a wealth-management trading product serving **100K+** clients and advisors, and architected Continuum, a multi-tenant Flink platform running **1,000+** tenant pipelines. Focused on correctness and reliability in distributed systems, including idempotent processing, exactly-once semantics, and schema evolution.

WORK EXPERIENCE

Senior Software Engineer

JP Morgan Chase

Aug '22 — Present

Plano, TX

- **Digital Wealth Management Trading Platform** | *Java, Spring Boot, Kafka, AWS ECS, Terraform, Datadog*
 - Own the backend API layer for a digital wealth-management trading platform used daily by **100K+** clients and financial advisors, serving real-time portfolio reads at **sub-200ms** latency.
 - Architected Spring Boot microservices across portfolio, goals, and investment domains, integrating synchronous REST APIs with asynchronous Kafka-driven investment triggers and goal-state transitions.
 - Guaranteed financial data correctness under retries and replays with idempotent Kafka consumers, explicit retry contracts, backpressure handling, and schema-versioned events.
 - Automated infrastructure and deployments with Terraform, Docker, and AWS ECS, making every service independently deployable and cutting environment setup and release cycle time by **40%**.
 - Defined backend guardrails (versioned API contracts, service boundaries, RBAC, production-readiness checks) with product, platform, and security teams, adopted across **10+** services; lead design reviews and mentor **3** engineers.
- **Continuum: Multi-Tenant Streaming Data Platform** | *Apache Flink, Kafka, AWS EKS, S3, Glue, Athena, Groovy*
 - Architected and delivered Continuum, a multi-tenant streaming data platform on AWS EKS running **1,000+** isolated tenant pipelines and processing millions of events per day.
 - Designed a Groovy-based declarative DSL that lets data engineering teams define pipelines as composable DAGs without writing Flink code, cutting pipeline onboarding from **weeks to hours**.
 - Built the Apache Flink execution engine: each tenant pipeline runs as an isolated job with **exactly-once** semantics, checkpoint-triggered Parquet writes to S3, and Glue Catalog registration for Athena queries.
 - Improved pipeline reliability with stateful processing, checkpointing, and automated failure recovery; eliminated Glue API throttling by batching partition registration.
 - Enforced tenant isolation via S3 prefixes, IAM policies; autoscaled EKS pods on per-tenant Kafka consumer lag.

Senior Software Engineer

AT&T

Aug '21 — Aug '22

Plano, TX

- **Cloud Migration - BILL PAY** | *Java, Spring Boot, Kafka, AWS, Terraform, Datadog*
 - Led migration of a legacy on-prem platform to AWS (ECS and Kubernetes) with no disruption to live operations, moving **15+** services onto Terraform-managed infrastructure.
 - Decomposed a monolith, where every release touched every component, into independently deployable Spring Boot microservices with clean REST contracts and per-service scaling.
 - Built Jenkins and GitHub Actions CI/CD pipelines with Terraform provisioning, enabling independent deployments and reducing release effort by **30%**.
 - Built observability with Datadog, CloudWatch, and Grafana dashboards and alerting, speeding incident detection and improving MTTR by **35%**.

Senior Software Engineer

Capital One

Jan '20 — Aug '21

Plano, TX

- **Auto Loans Backend Platform** | *Java, Spring Boot, Kafka, AWS, Terraform, Datadog*
 - Built Java Spring Boot microservices for Auto Loans origination, servicing, and underwriting APIs, keeping partner-facing REST APIs responsive under peak traffic.
 - Built Kafka pipelines for loan, payment, and underwriting events with strict ordering and delivery guarantees.
 - Raised event and batch throughput by **35%** under peak load using Java concurrency APIs and multithreading.
 - Secured PCI-scoped data with field-level encryption, tokenization, and RBAC across all API surfaces.

Software Engineer

Verizon

May '15 — Jan '20

Irving, TX

- **OPTIX Ordering Suite** | *Java, Spring, REST APIs, PCF, PostgreSQL*
 - Built REST APIs on Spring framework and integration modules for the OPTIX Ordering Suite, supporting FiOS order management and provisioning for thousands of call-center operators daily.
 - Reduced order-processing latency by **~25%** through SQL query tuning, caching, and asynchronous processing.
 - Migrated **20+** legacy SOAP endpoints to REST APIs, reducing service coupling across ordering integrations.

OPEN-SOURCE PROJECTS

StreamForge [Link](#)

- **YAML-to-Flink pipeline compiler**

- Compiles source | transform | sink YAML into Flink jobs with exactly-once semantics, resolving connectors and schema, and registering output in the Glue catalog. Java 17, Flink 1.18+, Docker.
- Designed a pluggable connector SPI for Kafka, S3, Iceberg, and Snowflake, so new sinks are added as modules rather than forks; includes a control plane, checkpoint-driven emission, and per-tenant isolation.

Flink for Dummies [Link](#)

- **Interactive guide to the Flink runtime**

- Four browser labs that port Flink's real key-group hashing (murmurHash, KeyGroupRangeAssignment), watermark rules, aligned and unaligned checkpoint barriers, and backpressure metrics. TypeScript, Docusaurus.
- 60-page textbook from events to runtime internals; each page covers the problem, mental model, what Flink does internally, how it fails, and what changes at scale.

SKILLS

Languages Java, Python, SQL, Groovy, Shell

Backend Microservices, REST APIs, Hibernate/JPA, JUnit, Mockito

Distributed Systems Kafka, Flink, Event-Driven Architecture, Multi-Tenancy, Idempotency, Exactly-Once Semantics

AWS ECS, EKS, S3, Glue, Athena, Lambda, IAM, Aurora/RDS, DynamoDB, API Gateway, CloudWatch

Data PostgreSQL, Cassandra, Redis, PySpark

Infrastructure Kubernetes, Docker, Terraform, Jenkins, GitHub Actions

Observability Datadog, CloudWatch, Prometheus, Grafana, Dynatrace, Spinnaker

CERTIFICATIONS

[AWS Certified Solutions Architect - Associate SAA-C03](#)

[HashiCorp Certified: Terraform Associate \(004\)](#)

[AWS Certified Developer Associate DVA-C02](#)

EDUCATION

Master of Science in Electrical Engineering, State University of New York

Aug '13 — May '15
New Paltz, New York, United States

- Specialized in embedded systems, algorithms, distributed computing, networking, and database management. Gained hands-on experience through Java-based projects and IoT applications, bridging hardware and software development.

Bachelor of Technology in Electrical Engineering, Kakatiya University

Aug '09 — May '13
Warangal, Telangana, India

- Built a solid foundation in electrical systems, programming, and applied mathematics, with a strong emphasis on analytical problem solving and system modeling.